

# Quiet Neurons: a BDH quietens on what it just learned, not on what is merely predictable

DataForge 2026, Pathway track · concept **Sparse Non-Negative Activations** · Nilay Toshniwal, solo · artifact [nilaymastaadmi.github.io/quiet-neurons](https://nilaymastaadmi.github.io/quiet-neurons) · code [github.com/nilaymastaadmi/quiet-neurons](https://github.com/nilaymastaadmi/quiet-neurons)

## THE DESIGN PRESSURE

A dense Transformer evaluates every feed-forward unit for every token, hard or not. Activation sparsity is the counter-proposal: fewer active units, cheaper inference, an active set small enough to name. But Transformer sparsity is already *emergent*, appearing during training rather than by design [3]; it needs a hard rectifier, because only that yields exact zeros [4]; and *which* units fire is input-dependent, not merely how many [5]. Sparsity by itself therefore distinguishes nothing. What distinguishes an architecture is what its sparsity is a function of.

## WHAT BDH CHANGES

Dragon Hatchling (BDH) [1] makes it structural. In the BDH-GPU formulation Pathway open-sources, every layer computes two ReLU'd vectors and multiplies them elementwise, and a product of two sparse non-negative vectors is sparser than either. Attention is the second change: query and key are the same tensor, the code asserting  $K$  is  $Q$ , so the score matrix is a causally-masked Gram matrix of neuron activations with **no softmax** in it. Attention is co-firing similarity, not a learned lookup — the paper's Hebbian reading, in one line of tensor code.

## THE CLAIM, AND THE CORRECTION IT NEEDED

Section 6.4 reports that "fewer neurons are active ... for more predictable input signals": at layer 2 of a 65,536-neuron model, 4.0–7.5% active while memorising a new word against about 2.5% while repeating it [1, Fig. 14]. We reproduced that protocol on three models trained here from the public code. It survives being shrunk by a factor of 32.

| Model (neurons / parameters) | Memorising | Repeating | Ratio    | Evidence class     |
|------------------------------|------------|-----------|----------|--------------------|
| ours, 2,048 / 397k           | 8.24%      | 5.76%     | 1.43×    | reproduced here    |
| ours, 8,192 / 3.15M          | 7.18%      | 3.64%     | 1.97×    | reproduced here    |
| ours, 16,384 / 6.30M         | 8.20%      | 4.38%     | 1.87×    | reproduced here    |
| Pathway, 65,536 [1]          | 4.0–7.5%   | ~2.5%     | 1.6–3.0× | developer-reported |

Each row is 1,280 sequences over five pinned samples, spread near 0.005. The same measurement then breaks the usual reading of it. Two positions in one sequence are predicted almost perfectly yet differ fourfold in activity: letter 11, inside the fixed warm-up, sits at **13.6%** active at 0.000 nats of surprise; letter 40, deep in the repeated word, at **3.5%** at 0.003 nats. The warm-up is identical in every training sequence, so it lives in the **weights**; the repeated word is new each sequence, so it can only come from the **context**. **The variable is where the knowledge lives**, not how predictable the text is, and the activation signature separates parametric from in-context memory. A learner falsifies this in under a minute by injecting an unpredictable letter mid-repetition: activity rises at that letter, +10% to +20% by word, and is unchanged to nine decimals at the letter before it, because the model reads only leftwards.

## EVIDENCE, LABELLED

**REPRODUCED HERE:** the three ratios above, the counterexample, and parity of the in-browser model against PyTorch (logit max difference 4.1e-5, eleven of twelve sparsity series bit-identical). **DEVELOPER-REPORTED, NOT REPRODUCED:** the Figure 14 band; 97.4% on Sudoku Extreme, which Pathway's own repository states comes from their internal implementation and does not reproduce from the open code; 1B-to-600B pretraining-scaling experiments; and BDH-CQ's 29.5% pass@2 on ARC-AGI-1 at \$0.0007 per task [2]. **NOT MODELLED:** BDH-CQ's internals, whose "dimensions, exact update rules, and implementation details remain proprietary" [2]. **BDH-CQ has no direct role in this concept**; inventing one would be worse than saying so.

## LIMITATIONS, MATURITY, AND WHAT WOULD SETTLE THEM

The effect is not uniform with depth: layer 0 runs *backwards* at all three sizes (0.82, 0.79, 0.86), and layer 3 is flat at the two smaller models and weakly positive (1.11×) at the largest. Scaling is **not monotonic** — 16,384 measures below 8,192 by roughly twenty times the sampling spread — and we published the opposite before the third model finished. One confound cannot be excluded: a single 4,000-step schedule cut by wall clock at 1,854, 1,917 and 2,309 steps, so the three stopped at different points on it. Final losses (0.176, 0.175, 0.172) say comparably trained, not controlled; three step-matched runs would settle it. These are toys, 0.4M to 6.3M parameters on a synthetic task, as the paper's own protocol is. Surprise and sparsity do *not* track per letter (Pearson 0.35, Spearman -0.05), which killed a stronger claim. And we count the elementwise product, the only quantity in the public code whose magnitude reaches the paper's band; the paper does not say which tensor it means, though that choice changes the size of the effect, not its direction, positive on all three candidates. **MATURITY:** for this narrow property BDH is credible and now outsider-checkable at small scale. The wider programme is earlier — headline benchmarks are developer-reported, the open repository ships no trained weights, and published comparisons stop at GPT-2 scale. Its interpretability claims are the verifiable ones today, and this is one.

[1] Kosowski, Uznański, Chorowski, Stamirowska, Bartoszkiewicz, *The Dragon Hatchling*, arXiv:2509.26507 (2025), §6.4 and Fig. 14. [2] Engdahl, Kosowski, Chorowski et al., *BDH-CQ*, arXiv:2608.09888 (2026). [3] Li, You, Bhojanapalli et al., *The Lazy Neuron Phenomenon*, arXiv:2210.06313 (2022), ICLR 2023. [4] Mirzadeh, Alizadeh, Mehta et al., *ReLU Strikes Back*, arXiv:2310.04564 (2023). [5] Liu, Wang, Dao et al., *Deja Vu*, arXiv:2310.17157 (2023), ICML 2023. — Reproduce every number: `python measure.py --ckpt checkpoints/bdh_n2048.pt --embd 64 --mult 32 --repeats 5`. Code and data MIT; bdh.py is Pathway's, unmodified, MIT.